

Zhihao Lin

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EDUCATION

Santa Clara University

Master of Science in Computer Science

GPA: 3.77/4.0; GRE: 334

Sep 2020 – Dec 2022

Santa Clara, CA

University at Buffalo, The State University of New York

Bachelor of Science in Computer Science

Cum Laude; Dean's List (2019)

Aug 2016 – Jun 2020

Buffalo, NY

INDUSTRY EXPERIENCE

Amazon Web Services — Amazon Connect

Software Development Engineer

Feb 2023 – Apr 2026

Seattle, WA

- Built and operated a large-scale metrics platform for cloud contact center as the service scaled from 10M+ to 20M+ daily customer interactions, supporting the computation, storage, validation, and API-based data vending to downstream consumers across distributed backend services.
- Owned and productionized an internal metrics validation service using AWS Lambda, DynamoDB, OpenSearch, and S3, expanding pre-launch validation coverage to over 90%, reducing customer complaints related to metric definitions by approximately 70%, and cutting investigation time for related support tickets from one day to two hours.
- Designed a GDPR-related data-deletion workflow across multiple regional OpenSearch clusters, using SQS-based scheduling with load-test-calibrated throttling, idempotent delete-by-query execution, and post-deletion verification scans to meet compliance SLAs without degrading cluster availability.
- Owned CI/CD pipelines, monitoring, and Tier-2 on-call operations for the platform, partnering with PMs and cross-functional engineering teams to launch metrics and resolve production incidents.
- Built retrieval-grounded metric onboarding workflows using DynamoDB and OpenSearch metadata, schemas, dependencies, and calculation logic to automate validation and debugging.
- Developed MCP-style tool-integration infrastructure exposing backend services for schema inspection, metric validation, and script execution, with structured validation guardrails for reliable LLM-driven workflows.
- Introduced the team's first AI-assisted SOP-to-workflow automation prototype, converting operational procedures into structured execution workflows and helping reduce the regional expansion timeline from two weeks to one week.

Amazon Web Services — Amazon Connect

Software Development Engineer Intern

Jun 2022 – Sep 2022

Seattle, WA

- Prototyped an early metrics validation service using AWS Lambda, DynamoDB, OpenSearch, and S3; presented it in an organization-wide demo and later productionized the service as a full-time engineer.

RESEARCH INTERESTS

Large language models and AI/ML systems, with interests in adaptive inference and generation, agentic and tool-using systems, retrieval and data systems, representation learning, and evaluation. I am particularly interested in building reliable LLM systems that combine model behavior with scalable data, retrieval, and tool-use infrastructure for interactive environments.

PUBLICATIONS

Read the Room Before Your Model Generates

2026

Muyao Kong, Zhihao Lin** — Accepted to AACL-IJCNLP 2026 Main Conference

A scenario-adaptive decoding framework that derives prompt-level sampling temperature from interpretable scenario representations learned from Transformer hidden states. *Equal contribution.

RESEARCH EXPERIENCE

Read the Room Before Your Model Generates

Jan 2026 – Present

Independent Researcher

Initial guidance from Prof. Haifeng Xu, University of Chicago; research conducted independently.

- Co-developed Scenario-Actor, a scenario-adaptive decoding framework that learns structured representations of communicative settings from Transformer hidden states and derives prompt-level sampling temperatures from their geometry.
- Co-designed and implemented Scenarize, an RNN-based encoder that maps intermediate Llama-3.1-8B-Instruct hidden states into 32-dimensional scenario embeddings using triplet-based representation learning.
- Co-designed Actout, an interpretable scenario-axis readout that maps scenario-embedding geometry to prompt-level decoding temperatures.
- Curated and GPT-4o-annotated 37K+ conversational prompts across 9 scenario labels and validated annotation quality against human raters.
- Evaluated Scenario-Actor on AgentSense, AlpacaEval, and MT-Bench, achieving the highest AgentSense goal-completion score and AlpacaEval LC-WR among evaluated decoding baselines; conducted representation-geometry and layer-sweep analyses to validate the learned scenario space.

Sentiment Analysis for Email Interaction and Fake News Research

Jan 2022 – Apr 2022

Research Assistant, Santa Clara University

Santa Clara, CA

Advised by Prof. Yi Fang.

- Contributed to an industry-collaborative NLP research project on sentiment analysis by reviewing related literature, participating in research discussions, and deploying experimental models on AWS EC2.
- Built a Chrome/Gmail sentiment-analysis plugin prototype that integrated NLP inference into email workflows for lightweight sentiment interpretation.
- Developed reusable research software that crawled web images associated with news reporters and used Amazon Rekognition to extract facial attributes, producing structured reporter data for downstream fake-news detection research; packaged the software for use by a PhD researcher.

SKILLS

Programming Languages: Python, Java, TypeScript, Bash

Machine Learning Frameworks & Libraries: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, Matplotlib

Systems & Development Tools: Linux, Git, Docker, Jupyter, LaTeX

Cloud Technologies: AWS Lambda, EC2, SageMaker AI, DynamoDB, OpenSearch, S3, SNS, SQS, CloudWatch, CloudFormation